

Subject Name:

Microwave Engineering & Antenna Theory

Subject Code: BEC701

Course Coordinator:

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Assistant Professor

Dept. of E&C, ATME

Module-1

Microwave Sources: Introduction, Gunn Diode (Text 2: 7.1,7.1.1,7.1.2)

Microwave transmission lines: Microwave frequencies, Microwave devices, Microwave systems. Transmission line equations and solutions, Reflection Coefficient and Transmission Coefficient. Standing wave and standing wave ratio. Smith chart, Single stub matching.

Text 2: 0.1, 0.2, 0.3, 3.1, 3.2, 3.3, 3.5, 3.6 (except double stub matching)

Module-2

A Closer Look at Methods and classes: Overloading methods, Using objects as parameters, Returning

Microwave Network Theory: Introduction, S matrix representation of multi-port networks (Text 1: 6.1, 6.3, 6.3.1, 6.3.2)

Microwave passive devices: Coaxial connectors and Adapters, Attenuators, Phase shifters, waveguide Tees, Magic Tee, Circulator, Isolator. (Text 1: 6.4.2, 6.4.14, 6.4.15, 6.4.16, 6.4.17 A, B)

Module-3

Strip Lines: Introduction, Microstrip lines, Parallel Strip lines (Text 2: 11.1,11.2)

Antenna Basics: Introduction, Basic Antenna Parameters, Patterns, Beam Area, Radiation Intensity, Beam efficiency, Directivity and Gain, Antenna Aperture Effective height, Bandwidth, Radio communication Link, Antenna Field Zones (Text 3: 2.1-2.7, 2.9-2.11, 2.13).

Teaching-Learning Process

Chalk and talk method, Power point presentation and videos.
RBT Level: L1, L2, L3

Module-4

Point sources and arrays: Introduction, Point Sources, Power patterns, Power theorem, Radiation Intensity, Arrays of 2 isotropic point sources, Pattern multiplication, Linear arrays of n Isotropic sources of equal amplitude and Spacing. (Text 3: 5.1-5.6, 5.9, 5.13)

Electric Dipole: Introduction, Short Electric dipole, Fields of a short dipole. Radiation resistance of a short dipole. Thin linear antenna (field analysis). (Text 3: 6.1-6.5)

Teaching-Learning Process

Chalk and Talk, PowerPoint Presentation
RBT Level: L1, L2, L3

Module-5

Loop and Horn antenna: Introduction: Small loop, Comparison of far fields of small loop and Short dipole. Radiation resistance of small loop, Horn Antennas, Rectangular antennas. (Text 3: 7.1, 7.2, 7.4, 7.6, 7.7, 7.8, 7.19, 7.20)

Antenna Types: The Helix geometry, Helix modes, Practical design consideration for mono-filar axial mode Helical Antenna, Yagi Uda array, Parabolic Reflector (Text 3: 8.3, 8.4, 8.5, 8.8, 9.5)

Teaching-Learning Process

Chalk and Talk, PowerPoint Presentation
RBT Level: L1, L2, L3

PRACTICAL COMPONENT OF IPCC	
Sl.No	Experiments
1	Study of characteristics of Magic Tee.
2	Coupling and Isolation characteristics of microstrip directional coupler.
3	Determination of power division of microstrip power divider.
4	Determination of resonance characteristics of microstrip ring resonator and computation of dielectric constant of the substrate.
5	Measurement of frequency, guide wavelength, power and attenuation in a microwave Test bench.
6	Study of characteristics of E plane Tee / H plane Tee.
7	To measure unknown impedance using Smith chart through test bench setup.
8	Measurement of VSWR and reflection coefficient and attenuation in a microwave test bench setup.
9	Obtain the radiation pattern of a Yagi-Uda Antenna array and calculate its directivity.
10	Calculate the aperture of a Dipole Antenna.
11	Obtain the near and far fields of a given antenna and compare the fields.
12	Obtain the bandwidth of a given Antenna.

Course Outcomes

At the end of the course the student will be able to:

1. Describe the use and advantages of microwave transmission
2. Analyze various parameters related to transmission lines.
3. Identify microwave devices for several applications.
4. Analyze various antenna parameters and their significance in building the RF system.
5. Identify various antenna configurations for suitable applications.

Suggested Learning Resources:

Text Books:

1. Microwave Engineering -Annapurna Das, Sisir K Das, TMH Publication, 2nd Edition, 2010.
2. Microwave Devices and Circuits – Samuel Y Liao, Pearson Education.
3. Antennas and Wave Propagation -John D Krauss, Ronald J Marhefka, Ahmad S Khan, 4th Edition, McGraw Hill Education, 2013.

Reference Books:

1. Microwave Engineering -David M Pozar, John Wiley India Pvt Ltd., Pvt Ltd., 3rd edition, 2008.
2. Microwave Engineering-Sushrut Das, Oxford Higher Education, 2nd Edn, 2015.
3. Antennas and Wave Propagation- Harish and Sachidananda, Oxford University Press, 2007.

Web links and Video Lectures (e-Resources):

- https://www.tutorialspoint.com/antenna_theory/antenna_theory_horn.html
- <http://www.antenna-theory.com/antennas/smallLoop.php>

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Quizzes, Seminars

Module 1

Microwave Sources and Microwave Transmission Lines

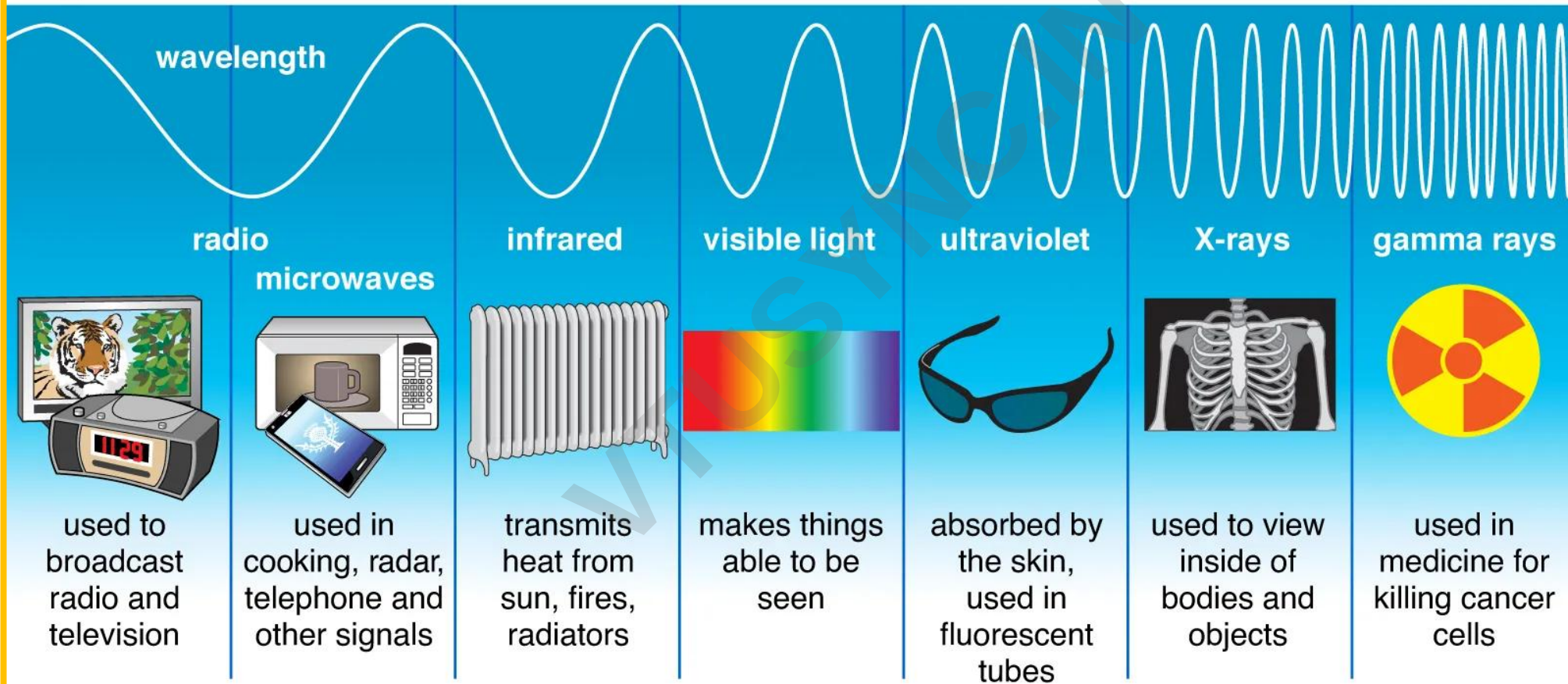
Module-1

Microwave Sources: Introduction, Gunn Diode (Text 2: 7.1,7.1.1,7.1.2)

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Types of Electromagnetic Radiation



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Microwaves

- Microwaves are defined as electromagnetic radiations with a frequency ranging between **300 MHz to 300 GHz**. In contrast, the wavelength ranges from 1 mm to around 30 cm.
- Microwave radiation is commonly referred to as microwaves.
- They fall between infrared radiation and radio waves in the electromagnetic spectrum.

A transferred electron device (TED),

- A transferred electron device (TED), is a form of diode, a two-terminal semiconductor electronic component, *with negative differential resistance*, used in high-frequency electronics.

Negative Resistance:

- Negative Resistance (NR) is a property of some electrical circuits and devices in which an increase in voltage across the device's terminals results in a decrease in electric current through it.

Difference between microwave transistor and TED(Transfer Electronic Device).

Microwave Transistor	TED (Transfer Electronic Device)
It consists of junctions or gates.	It is not having any junction or gate, it is a bulk type of device
It is fabricated from elemental semiconductor such as Ge and Si	It is fabricated from compound semiconductor such as Gallium Arsenide (GaAs), Indium Phosphide (Inp), and Cadmium Telluride (CdTe)
It operates with low energy "Warm" electrons	It operates with high energy "hot" electrons.
Advantages: Low cost, low power supply, small size	Advantages: Low noise, high gain, low power, reliable, lighter in weight

GUNN-EFFECT DIODES - GaAs DIODE

- Gunn-effect diodes are named after [J. B. Gunn](#), who discovered it in 1963.
- **Gunn Effect:** A periodic fluctuations of current passing through the n-type gallium arsenide (GaAs) specimen when the applied voltage exceeded a certain critical value.

Periodic Table of the Elements

1 1IA 11A	2 IIA 2A											13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 H Hydrogen 1.0079																	2 He Helium 4.00260
3 Li Lithium 6.941	4 Be Beryllium 9.01218											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.00674	8 O Oxygen 15.9994	9 F Fluorine 18.998403	10 Ne Neon 20.1797
11 Na Sodium 22.989768	12 Mg Magnesium 24.305	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 Al Aluminum 26.981539	14 Si Silicon 28.0855	15 P Phosphorus 30.973762	16 S Sulfur 32.066	17 Cl Chlorine 35.4527	18 Ar Argon 39.948
19 K Potassium 39.0983	20 Ca Calcium 40.078	21 Sc Scandium 44.95591	22 Ti Titanium 47.88	23 V Vanadium 50.9415	24 Cr Chromium 51.9961	25 Mn Manganese 54.938	26 Fe Iron 55.847	27 Co Cobalt 58.9332	28 Ni Nickel 58.6934	29 Cu Copper 63.546	30 Zn Zinc 65.39	31 Ga Gallium 69.723	32 Ge Germanium 72.64	33 As Arsenic 74.92159	34 Se Selenium 78.96	35 Br Bromine 79.904	36 Kr Krypton 83.80
37 Rb Rubidium 85.4678	38 Sr Strontium 87.62	39 Y Yttrium 88.90585	40 Zr Zirconium 91.224	41 Nb Niobium 92.90638	42 Mo Molybdenum 95.94	43 Tc Technetium 98.9072	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.9055	46 Pd Palladium 106.42	47 Ag Silver 107.8682	48 Cd Cadmium 112.411	49 In Indium 114.818	50 Sn Tin 118.71	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.90447	54 Xe Xenon 131.29
55 Cs Cesium 132.90543	56 Ba Barium 137.327	57-71 Lanthanide Series	72 Hf Hafnium 178.49	73 Ta Tantalum 180.9479	74 W Tungsten 183.85	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.9665	80 Hg Mercury 200.59	81 Tl Thallium 204.3833	82 Pb Lead 207.2	83 Bi Bismuth 208.98037	84 Po Polonium [209]	85 At Astatine [210]	86 Rn Radon 222.0176
87 Fr Francium 223.0197	88 Ra Radium 226.0254	89-103 Actinide Series	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [268]	110 Ds Darmstadtium [269]	111 Rg Roentgenium [272]	112 Cn Copernicium [277]	113 Uut Ununtrium unknown	114 Fl Flerovium [289]	115 Uup Ununpentium unknown	116 Lv Livermorium [293]	117 Uus Ununseptium unknown	118 Uuo Ununoctium unknown
			57 La Lanthanum 138.9055	58 Ce Cerium 140.115	59 Pr Praseodymium 140.90765	60 Nd Neodymium 144.24	61 Pm Promethium 144.9127	62 Sm Samarium 150.36	63 Eu Europium 151.9655	64 Gd Gadolinium 157.25	65 Tb Terbium 158.92534	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93032	68 Er Erbium 167.26	69 Tm Thulium 168.93421	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.967
			89 Ac Actinium 227.0278	90 Th Thorium 232.0381	91 Pa Protactinium 231.03588	92 U Uranium 238.0289	93 Np Neptunium 237.0482	94 Pu Plutonium 244.0642	95 Am Americium 243.0614	96 Cm Curium 247.0703	97 Bk Berkelium 247.0703	98 Cf Californium 251.0796	99 Es Einsteinium [254]	100 Fm Fermium 257.0951	101 Md Mendelevium 258.1	102 No Nobelium 259.1009	103 Lr Lawrencium [262]
			Alkali Metals	Alkaline Earths	Transition Metals	Basic Metals	Semi-Metals	Nonmetals	Halogens	Noble Gases	Lanthanides	Actinides					

A schematic diagram of a uniform n-type GaAs diode with ohmic contacts at the end surfaces is shown in Fig.

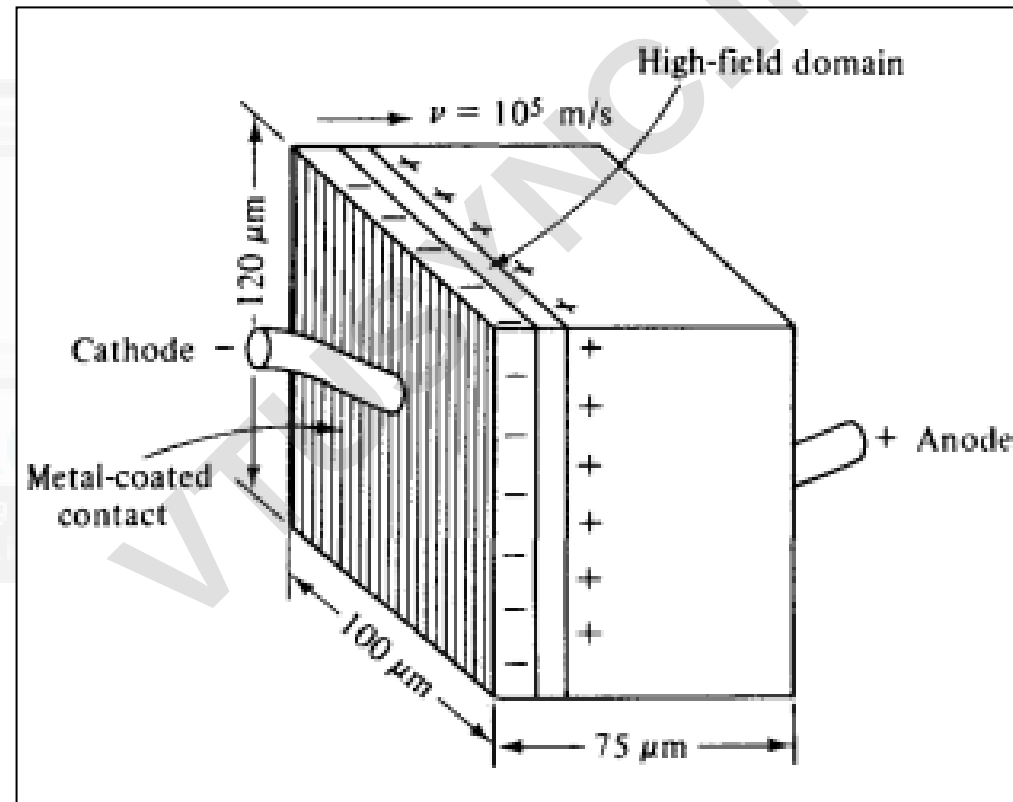


Fig: Schematic diagram for n-type GaAs diode.

- Gunn-effect diodes are named after J. B. Gunn, who in 1963 discovered a periodic fluctuations of current passing through the n-type gallium arsenide (GaAs) specimen when the applied voltage exceeded a certain critical value (2000-4000 V/cm).
- The frequency of oscillations was determined by specimen and not by the external circuit.
- The period and oscillations was usually inversely proportional to specimen length and is closely equal to *transit time* of electrons between electrodes. The drift velocity is over 10^5 m/sec

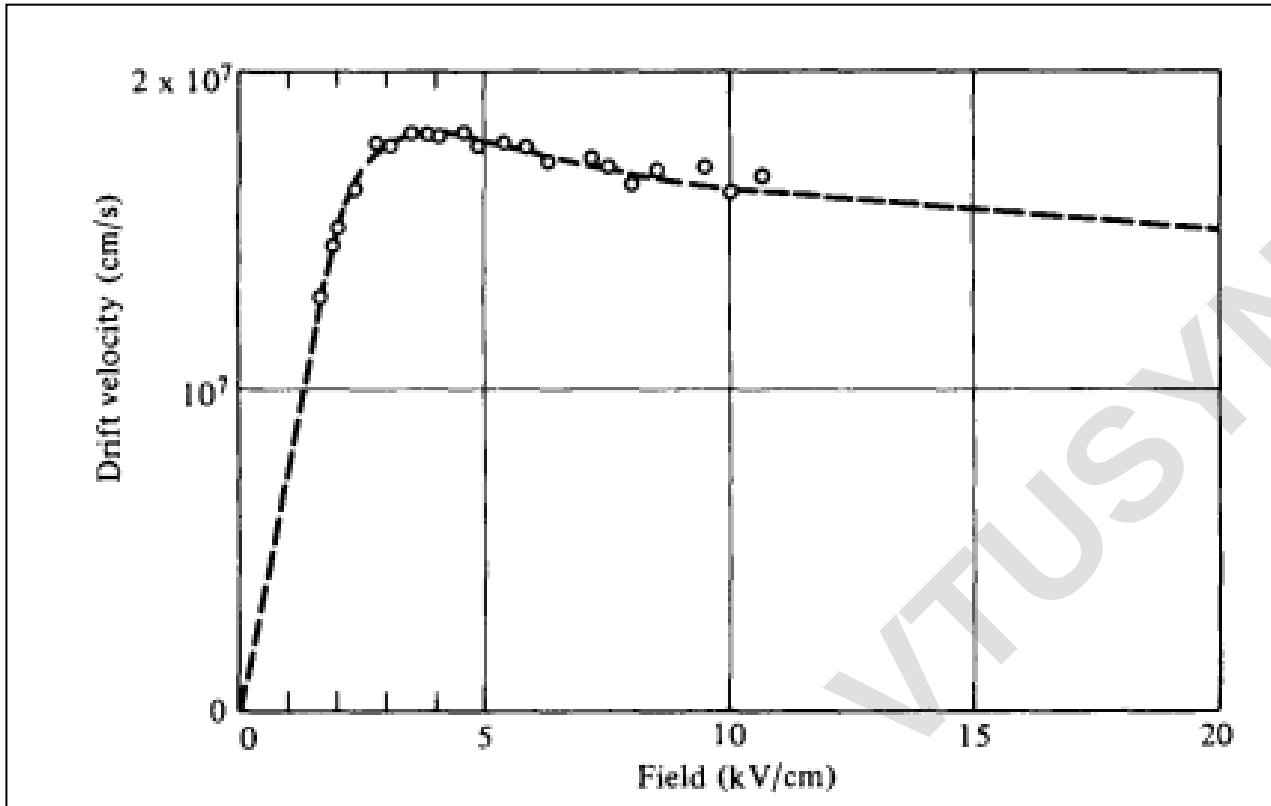


Fig: Drift velocity of electrons in n-type GaAs versus electric field

- The Drift velocity is linearly increased from zero to a maximum when the electric field is varied from zero to a threshold value.
- When the electric field is beyond the threshold value of 3000 V/cm for the n-type GaAs, the drift velocity is decreased and the diode exhibits negative resistance.
- The threshold electric field E_{th} is given by

$$E_{th} = \frac{V}{L}$$

The current fluctuations are shown in Fig

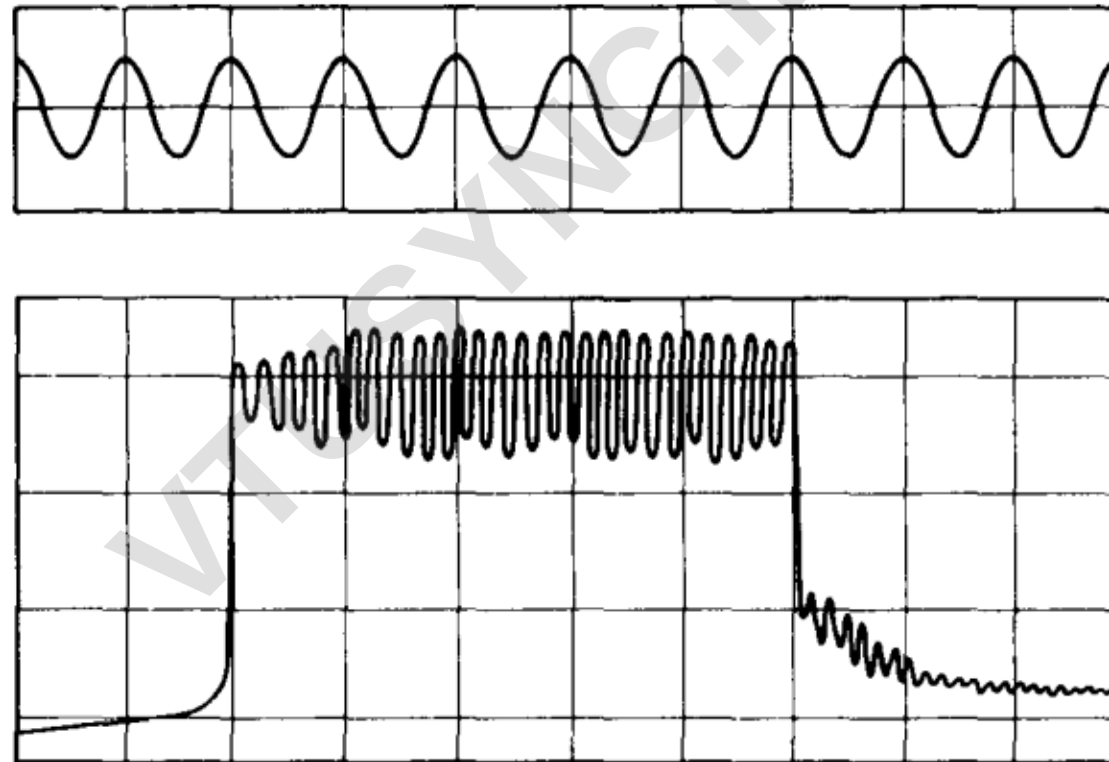


Fig: Current waveform of n-type GaAs

Problem:

The electric field distribution within a specimen of n-type GaAs of length $L = 210 \mu\text{m}$ and cross-sectional area $3.5 \times 10^{-3} \text{ cm}^2$ with a low-field resistance of 16Ω . Current instabilities occurred at specimen voltages is above 59 V .

Calculate the threshold electric field E_{th}

$$E_{\text{th}} = \frac{V}{L} = \frac{59}{210 \times 10^{-6} \times 10^2} = 2810 \text{ volts/cm}$$

Module-1

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Module 1: Chapter 2

Microwave Transmission Lines

- Microwave Frequencies ✓
- Microwave devices
- Microwave Systems
- Transmission Line equations and solutions
- Reflection Coefficient and Transmission Coefficient
- Standing Wave and Standing Wave Ratio,
- Smith Chart
- Single Stub matching

❏ Microwave Frequencies

- The term microwave frequencies is generally used for those wavelengths measured in centimeters, roughly from 1 mm to 30 cm (1 to 300 GHz)
- Microwave really indicates the wavelengths in micron ranges.

TABLE 0-1 U.S. MILITARY MICROWAVE BANDS

Designation	Frequency range in gigahertz
P band	0.225– 0.390
L band	0.390– 1.550
S band	1.550– 3.900
C band	3.900– 6.200
X band	6.200– 10.900
K band	10.900– 36.000
Q band	36.000– 46.000
V band	46.000– 56.000
W band	56.000–100.000

TABLE 0-2 U.S. NEW MILITARY MICROWAVE BANDS

Designation	Frequency range in gigahertz	Designation	Frequency range in gigahertz
A band	0.100–0.250	H band	6.000– 8.000
B band	0.250–0.500	I band	8.000– 10.000
C band	0.500–1.000	J band	10.000– 20.000
D band	1.000–2.000	K band	20.000– 40.000
E band	2.000–3.000	L band	40.000– 60.000
F band	3.000–4.000	M band	60.000–100.000
G band	4.000–6.000		

TABLE 0-3 IEEE MICROWAVE FREQUENCY BANDS

Designation	Frequency range in gigahertz	
HF	0.003–	0.030
VHF	0.030–	0.300
UHF	0.300–	1.000
L band	1.000–	2.000
S band	2.000–	4.000
C band	4.000–	8.000
X band	8.000–	12.000
Ku band	12.000–	18.000
K band	18.000–	27.000
Ka band	27.000–	40.000
Millimeter	40.000–	300.000
Submillimeter	>300.000	

Module 1: Chapter 2

Microwave Transmission Lines

- Microwave Frequencies ✓
- Microwave devices ✓
- Microwave Systems
- Transmission Line equations and solutions
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❏ Microwave Devices:

- Late 1930s: Vacuum Tubes
- In 1935: A. A. Heil and O. Heil - Lumped tuned circuit
- In 1939: W. C. Hahn and G. F. Metcalf- Velocity Modulation for microwave tubes
- In 1939: R.H. Varian and S. F. Varian – Two Cavity klystron amplifier
- In 1944: R. Kompfner – Helix type travelling-wave tube (TWT)
- In 1963: J.B. Gunn – “Gunn Effect”
- In the past two decades, however, microwave solid-state devices such as Tunnel Diodes, Gunn Diodes, Transferred Electron Devices (TEDs), Avalanche transit Time Devices have been developed.

Microwave Devices

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Module 1: Chapter 2

Microwave Transmission Lines

- Microwave Frequencies ✓
- Microwave devices ✓
- Microwave Systems ✓
- Transmission Line equations and solutions
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❑ Microwave Systems

Jan 2020

- A Microwave system normally consists of a transmitter subsystem, including a microwave oscillator, waveguides, and a transmitting antenna, and a receiver subsystem that includes a receiving antenna, transmission line or waveguide, a microwave amplifier, and a receiver.

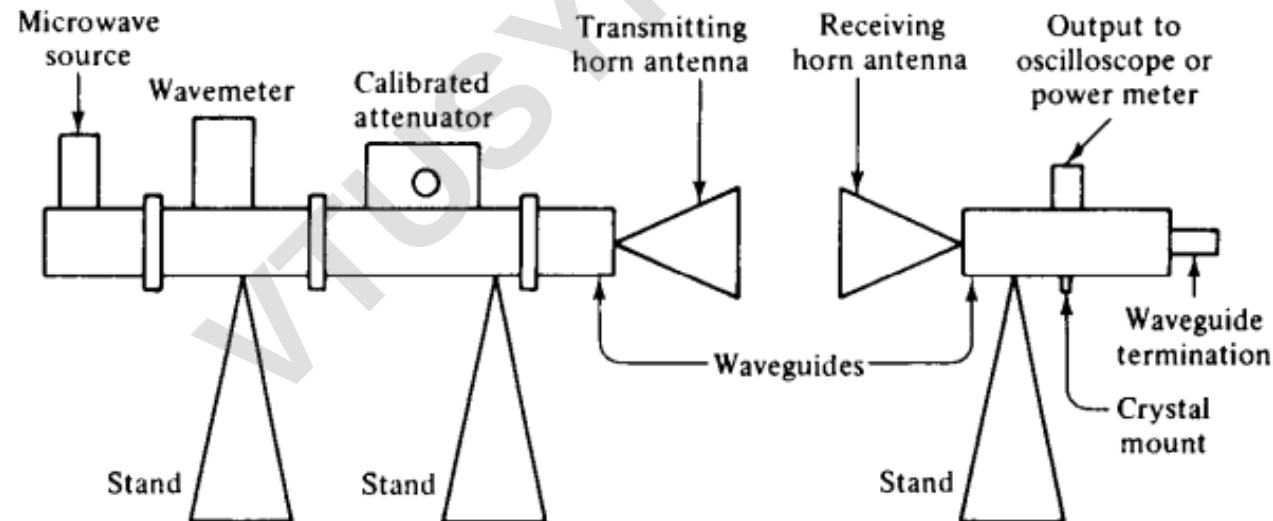
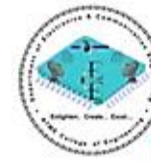


Fig: Microwave System



Thank You

